

claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-a3210bdb69f01fd58.jsonl`  
- SHA-256: `24f6ab5bb71753d215ea080e0f558140a00d8a6f42901d8122ae314417ea0859`  
- Source modified: `2026-06-16T03:20:05+00:00`  
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- project: `wf_3d536535-241`  
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

Transcript

1. user (2026-06-16T03:19:22.234Z)

Source Extractor

Research question: "Research question: **What concrete, recent (2022?2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 ? ~16 golden videos)?** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):

- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96?99% per-frame visibility).

- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.

- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval?serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best *recent* methods for (a) turning a dense per-frame signal into well-segmented actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMM/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k,

edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6716 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6716 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RL00CV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 375 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible."

Fetch and extract key claims from this source:

****URL:**** <https://arxiv.org/html/2501.11124v2/>

****Title:**** NoCo: Rethinking Pseudo-Label Guided Learning for WS-TAL from a Noise-Correction Perspective (AAAI 2025)

****Found via:**** Small-data / weak / timestamp-supervised TAL + active learning search

Task

1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
 - be a concrete, checkable statement (not vague generalities)
 - include a direct quote from the source as support
 - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-06-16T03:19:24.202Z)

I'll fetch the source and extract the relevant claims.

3. user (2026-06-16T03:19:41.759Z)

NoCo: Noise Correction for Weakly-Supervised Temporal Action Localization

1. Publication Venue and Date

The paper appears on arXiv as arXiv:2501.11124v2, dated January 2025.

2. Core Problem Addressed

NoCo identifies three types of noise in pseudo-labels generated by weakly-supervised models:

> "Issues with noisy labels include:(1) inaccurate boundary localization; (2) undetected short action clips; (3) multiple adjacent segments incorrectly detected as one segment."

The authors argue that "the noise in pseudo-labels would interfere with the learning of fully-supervised detection head, leading to significant performance leakage."

3. Named Components/Methods

NoCo consists of:

- **Context-Aware Label Augmentation (CALA)**: Refines instance boundaries through weighted averaging of neighboring instances
- **Ambiguous Instance Correction (AIC)**: Addresses "many-to-one" problems by mining high IoU instances
- **Missing Instance Compensation (MIC)**: Solves short-action omissions by identifying compensation instances with low IoU ratios
- **High-quality Pseudo-label Mining (HPM) Loss**: Assigns adaptive weights to training instances

The framework operates in two stages: pseudo-label generation with CALA, then online correction via teacher-student training.

4. Quantitative Results

THUMOS14:

- Achieves 50.9% average mAP (6.1% improvement over CO2-Net baseline)
- Surpasses previous state-of-the-art PivoTAL
- 3x faster inference speed

ActivityNet v1.2:

- Achieves 30.7% mAP (4.2% improvement over ASM-Loc)
- Outperforms prior state-of-the-art methods

Ablation studies show cumulative gains: CALA (+1.3%), Teacher-Student framework (+2.0%), AIC (+2.2%), MIC (+1.8%), combined AIC/MIC (+4.1%), HPM (+0.2%).

5. How It Handles Label Noise

The framework uses two-stage correction:

Stage 1: CALA applies "weighted aggregation of associated action instances" to correct boundaries before downstream training.

Stage 2: An "online-revised teacher-student framework" leverages "weighted aggregation of multiple historical student models" to generate stable correction signals, progressively updating pseudo-labels during training.

6. Pseudo-Label Quality and Training Strategy

> "our approach introduces several key innovations. Firstly, to generate robust and stable pseudo-label compensation information, our teacher model employs weighted aggregation of multiple historical student models."

Importantly: "the inference process of NoCo does not depend on the WTAL base model. In this paper, we build a student model based on the efficient TAL model TriDet, and during the inference phase, only the student model needs to be run."

4. assistant (2026-06-16T03:19:44.820Z)

This is a primary research paper (arXiv, AAAI 2025) directly relevant to the small-data / weakly-supervised TAL and over-segmentation dimensions. Let me extract the key claims.

5. user (2026-06-16T03:19:54.363Z)

Structured output provided successfully

6. assistant (2026-06-16T03:20:04.977Z)

Extracted 5 falsifiable claims from the NoCo paper (arXiv:2501.11124v2, January 2025, AAAI 2025). Source quality: primary research.

Key relevance to the research question: NoCo directly addresses the over-merge pathology (Dimension 1 ? "multiple adjacent segments incorrectly detected as one segment") via its named AIC/MIC correction modules, and operates in the weakly-supervised + pseudo-label-noise regime (Dimension 3). The teacher-student noise-correction framing and TriDet-based decoupled student are relevant to small-data and local-GPU constraints. One caveat for the parent agent: results are on THUMOS14/ActivityNet (multi-class video TAL), not 1-D trajectory data, so transfer to the shuttle-trajectory + cue setting is a research bet rather than a drop-in win.